Metals Market Index MMI

MMi Dashboard

Iron Ore Price Indices

Iron Ore Port Stock (FOT Qingdao)
IOPI62 62% Fe Fines RMB/t

858
16 1.90%
Dec 30th, 2022

Iron Ore Port Stock (FOT Qingdao)
IOPI65 65% Fe Fines RMB/t

970
16.00 1.68%
Dec 30th, 2022

Iron Port Stock (FOT Qingdao)
IOPI58 58% Fe Fines RMB/t

732
0 0.00%
Dec 30th, 2022

Iron Ore Seaborne (CFR Qingdao)
IOSI62 62% Fe Fines USD/dmt

113.35
0.00 0.00%
Dec 30th, 2022

Iron Ore Seaborne (CFR Qingdao)
IOSI65 65% Fe Fines USD/dmt

124.55
-2.35 -1.85%
Dec 30th, 2022

Iron Ore Port Stock (FOT Qingdao)
IOPLI 62.5% Fe Lump RMB/t

935
10 1.08%
Week Ending Dec 30th, 2022

Exchange Traded Contracts

DCE Iron Ore 62% Fines
I2305 (Jan) RMB/t (3pm close)

863.00
18.00 2.13%
Dec 30th, 2022 (3pm close)

SGX Iron Ore (CFR Qingdao) 62% Fe Fines
January 23 USD/dmt

117.30
2.50 2.18%
Dec 30th, 2022 (5.30 pm Print)

SHFE Rebar
RB2305 (Jan) RMB/t

4105
57 1.41%
Dec 30th, 2022 (3pm close)

Freight Rates

C3, Tubarao - Qingdao USD/t

HOLIDAY

Dec 29th, 2022

C5, W. Australia - Qingdao USD/t

HOLIDAY

Dec 29th, 2022

Steel Price

Steel Rebar (China Domestic) RMB/t

4100
60 1.49%
Week Ending Dec 30th, 2022

Inventory Levels

Iron Ore Inventory at Chinese Ports (35)
million tonnes

128.47
-1.54 -1.18%
Week Ending Dec 30th, 2022

Steel Inventory in China
million tonnes

9.36
0.45 5.08%
Week Ending Dec 30th, 2022

Steel Price

Steel HRC (China Domestic) RMB/t

4180
100 2.45%
Week Ending Dec 30th, 2022

IRON ORE PORT STOCK INDEX (IOPI)

Dec 30th, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOPI62	62% Fe Fines	858	16	1.9%	764	862	643	1083	114.63	2.45	2.2%	101.36	120.11	82.29	159.06
IOPI58	58% Fe Fines	732	0	0.0%	682	732	604	907	98.30	0.21	0.2%	91.00	102.16	77.29	134.15
IOPI65	65% Fe Fines	970	16	1.7%	893	1017	811	1265	130.10	2.48	1.9%	118.94	142.42	108.78	187.25

IRON ORE SEABORNE INDEX (IOSI)

Dec 30th, 2022		CFR Qingdao, USD/dry tonne						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOSI62	62% Fe Fines	113.35	0.00	0.00%	102.15	122.26	83.90	159.45
IOSI65	65% Fe Fines	124.55	-2.35	-1.85%	112.12	140.19	94.45	185.82

MARKET COMMENTARY

DCE iron ore futures rose by 2.8% today, the main contract closed 863.The traders' willingness to ship is general.The steel mills are wait and see.The overall trading sentiment of the market is general.PBF at Shandong port deal847-855yuan/mt,increase 25 yuan/mt; PBF at Tangshan port dealt 854-870 yuan/mt,increase 15-18 yuan/mt. As of December 30, the inventory of 35 ports tracked by SMM totaled 128.47 million tons, down 1.54 million tons from last week and 23.34 million tons from the same period last year. The average daily port dredging volume of imported ore in this period increased by 30000 tons to 2.957 million tons month on month. This week, the profits of the steel plant were repaired slightly, and some steel plants' blast furnaces were prepared to resume production, which, combined with the small replenishment of warehouses in the early New Year's Day, made the port more active in dredging. The new year has a "good start". It is expected that in January 2023, molten iron production will increase significantly, supporting iron ore demand again. At present, the price of raw materials continues to rise, and the steel mills may jointly expect the coke purchase price to rise and fall, so as to repair profits and improve

IRON ORE PORT LUMP INDEX (IOPLI)

Week Ending Dec 30th, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [3]						
Index	Fe Content	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]
IOPLI62	62.5% Fe Lump	935	10	1.1%	879	1043	770	1405	119.96	1.42	1.20%	112.19	140.23	94.72	199.13

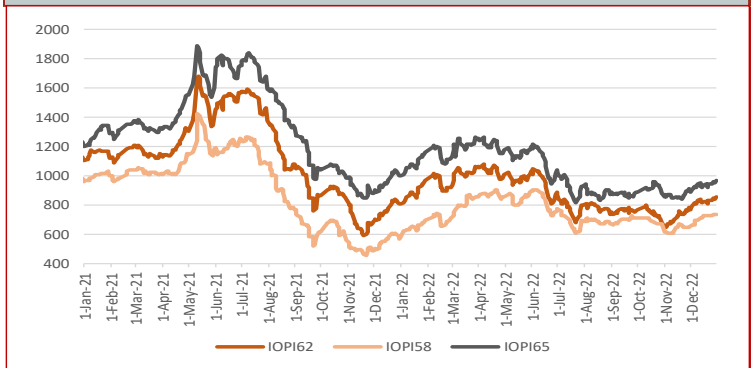
IRON ORE DOMESTIC CONCENTRATE SPOT PRICE ASSESSMENTS AND COMPOSITE INDEX

Week Ending Dec 30th, 2022				RMB/tonne (excluding tax) [3]				USD/tonne (excluding tax) [3]			
Province	Region	Product	Basis	This week	Change %	Low [2]	High [2]	This week	Change %	Low [2]	High [2]
Hebei	Hanxing	66% Fe Concentrate	Dry	1003	0.5%	779	1645	143.91	0.58%	110.31	255.69
Hebei	Qian'an	65% Fe Concentrate	Dry	1050	2.9%	780	1630	150.65	3.03%	110.51	251.57
Liaoning	Anshan	65% Fe Concentrate	Wet	740	0.0%	620	1310	106.17	0.08%	87.40	202.32
Shandong	Zibo	65% Fe Concentrate	Dry	1026	0.5%	800	1752	147.21	0.57%	117.19	272.32

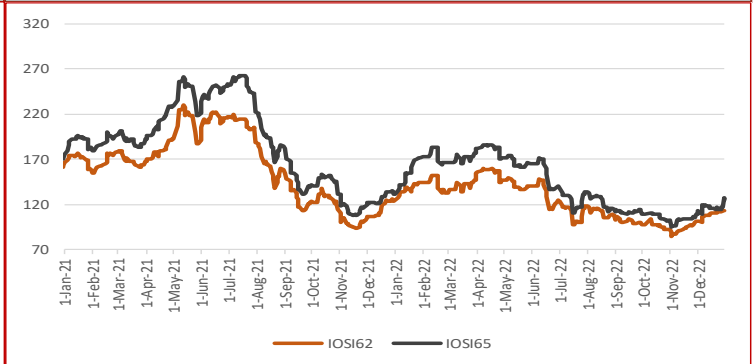
Week Ending Dec 30th, 2022	This week	Change %	Low [2]	High [2]
China Mines Concentrate Composite Index RMB/WT	905.49	0.95%	706.36	1511.22

[1] Exchange rate applied: RMB/USD = 6.7522 [2] Last 12 months
[3] Weekly exchange rate applied: RMB/USD =6.9596

IRON ORE PORT INDEX, FOT QINGDAO (RMB/WT)



IRON ORE SEABORNE INDEX, CFR QINGDAO (USD/DMT)



IRON ORE PORT STOCK INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Dec 30th, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne						
Index	Fe Content	September	October	November	December	MTD	QTD	YTD	September	October	November	December	MTD	QTD	YTD
IOPI62	62% Fe Fines	759	736	714	822	764	765	862	101.14	95.56	92.18	109.43	101.36	102.14	120.11
IOPI58	58% Fe Fines	697	683	635	711	682	682	732	93.56	89.34	82.39	95.17	91.00	91.57	102.16
IOPI65	65% Fe Fines	875	918	865	935	893	896	1017	117.19	120.08	112.37	124.90	118.94	120.24	142.42

IRON ORE SEABORNE INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Dec 30th, 2022		CFR Qingdao, USD/dry tonne						
Index	Fe Content	September	October	November	December	MTD	QTD	YTD
IOSI62	62% Fe Fines	101.12	95.80	93.05	108.92	102.15	103.41	122.26
IOSI65	65% Fe Fines	111.57	106.70	102.40	116.83	112.12	114.12	140.19

FREIGHT RATES

Dec 29th, 2022		FREIGHT RATES - DRY BULK US$/wet tonne			
Route	Designation	Change	Change %	Low [2]	High [2]
W. Australia - Qingdao	C5	HOLIDAY			
Tubarao - Qingdao	C3				

IRON ORE PORT LUMP INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Week Ending Dec 30th, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	September	October	November	December	MTD	QTD	YTD	October	October	November	December	MTD	QTD	YTD
IOPLI62	62.5% Fe Lump	873	877	811	922	879	884	1043	112.75	109.81	100.83	117.06	112.19	113.76	140.23

IRON ORE INDEX PREMIUMS/DISCOUNTS

Dec 30th, 2022		PORT STOCK INDEX (RMB/WT)	
Index	Fe Content	Spread to IOPI62	% Spread to IOPI62
IOPI58	58% Fe Fines	-126	-14.69%
IOPI65	65% Fe Fines	112	13.05%

Dec 30th, 2022		SEABORNE INDEX (USD/DMT)	
Index	Fe Content	Spread to IOSI62	% Spread to IOSI62
IOSI65	65% Fe Fines	11.20	9.88%

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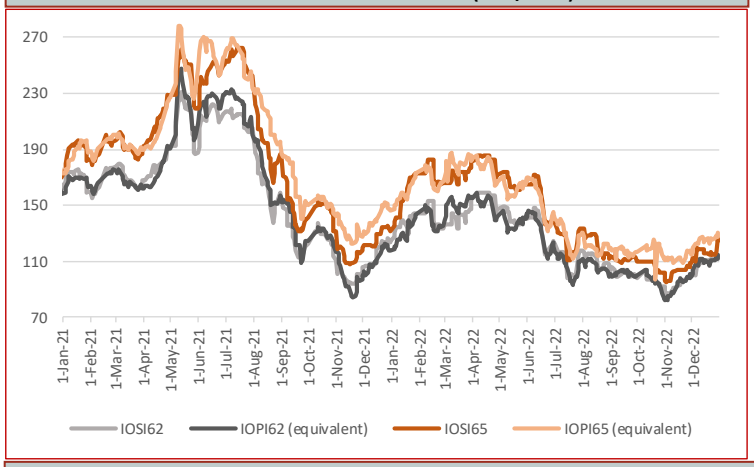
IRON ORE INDEX COMPARISONS (USD/DMT)



CHINA DOMESTIC COMPOSITE MINES INDEX (RMB/Wet Tonne, including VAT)

IRON ORE BRAND SPOT PRICE ASSESMENTS

Dec 30th, 2022	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI62
Roy Hill	801	16	-57
SIMEC Fines	730	16	-128
PB Fines	826	15	-32
Newman Fines	827	16	-31
MAC Fines	808	16	-50
Jimblebar Blended Fines	723	16	-135
Carajas Fines	1038	16	180
Brazilian SSF	822	17	-36
Brazilian Blend Fines	840	16	-18
RTX Fines	741	16	-117
West Pilbara Fines	769	16	-89

Dec 30th, 2022	SEABORNE INDEX (USD/DMT)		
	Price	Change	Diff to IOSI62
Roy Hill	108.85	0.00	-4.50
SIMEC Fines	105.35	0.00	-8.00
PB Fines	109.60	0.00	-3.75
Newman Fines	112.50	0.00	-0.85
MAC Fines	109.60	0.00	-3.75
Jimblebar Blended Fines	102.00	0.00	-11.35
Carajas Fines	142.90	0.00	29.55
Brazilian SSF	117.10	0.00	3.75
Brazilian Blend Fines	118.75	0.00	5.40
RTX Fines	103.25	0.00	-10.10
West Pilbara Fines	107.60	0.00	-5.75

Dec 30th, 2022	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI58
SSF	732	0	0
FMG Blended Fines	741	0	9
Robe River	742	0	10
Western Fines	744	0	12
Atlas Fines	739	0	7
Yandi	725	0	-7

IRON ORE INDEX NORMALISATION DIFFERENTIALS

Port Stock Index Product Differentials (RMB/wet tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	34.00	3.00
	High Grade Fe 63 - 64%	40.00	-2.00
	High Grade Fe 64 - 65%	40.00	-2.00
	High Grade Fe 65 - 65.5%	40.00	-2.00
	Low Grade Fe	23.00	0.00
1% Alumina	High Fe Grade Al <2.25%	83.00	0.00
	High Fe Grade Al 2.25-4%	49.00	-5.00
	Low Fe Grade Al <2.25%	22.00	-5.00
	Low Fe Grade Al 2.25-4%	23.00	-3.00
1% Silica	High Fe Grade Si <4%	43.00	-1.00
	High Fe Grade Si 4-6.5%	33.00	0.00
	Low Fe Grade	20.00	-2.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	28.00	0.00
	High Fe Grade 0.115%<P<0.15%	28.00	0.00
	Low Fe Grade 0.09<P<0.1%	5.00	0.00

Seaborne Index Product Differentials (USD/dry tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	2.25	-0.25
	High Grade Fe 63 - 64%	2.25	0.25
	High Grade Fe 64 - 65%	2.25	0.25
	High Grade Fe 65 - 65.5%	2.25	0.25
1% Alumina	High Fe Grade Al <2.25%	3.75	0.25
	High Fe Grade Al 2.25-4%	0.25	0.00
1% Silica	High Fe Grade Si <4%	2.50	0.25
	High Fe Grade Si 4 - 6.5%	0.50	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	2.00	0.00
	High Fe Grade 0.115%<P<0.15%	5.25	0.00

Port Stock Price Differentials to Qingdao Port for PB Fines (RMB/wet tonne)

Port	Value	Change	Port	Value	Change	Port	Value	Change	Port	Value	Change
Bayuquan	-100.00	0.00	Fangcheng	-25.00	0.00	Lanshan	0.00	0.00	Rizhao	0.00	0.00
Beilun	0.00	0.00	Jiangyin	-30.00	0.00	Lianyungang	0.00	0.00	Shekou	0.00	0.00
Caofeidian	-10.00	0.00	Jingtang	-20.00	0.00	Majishan	0.00	0.00	Taicang	-30.00	0.00
Dalian	0.00	0.00	Lanqiao	0.00	0.00	Qingdao	0.00	0.00	Tianjin	-10.00	0.00

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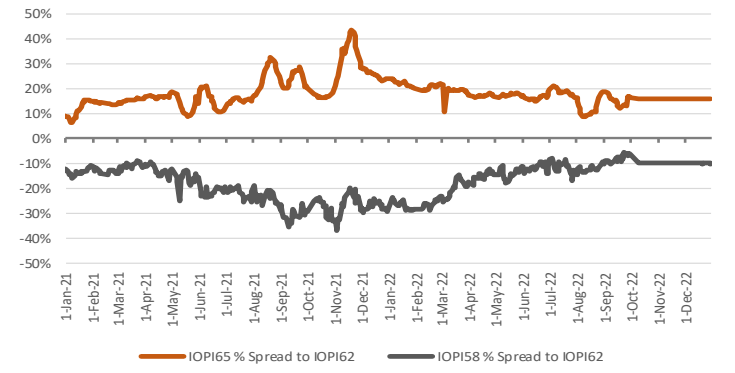
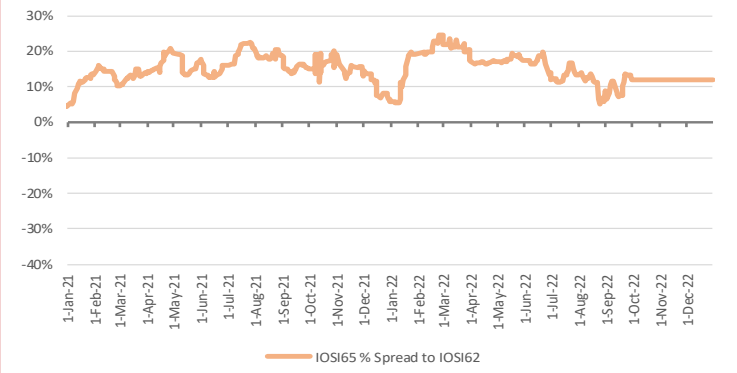
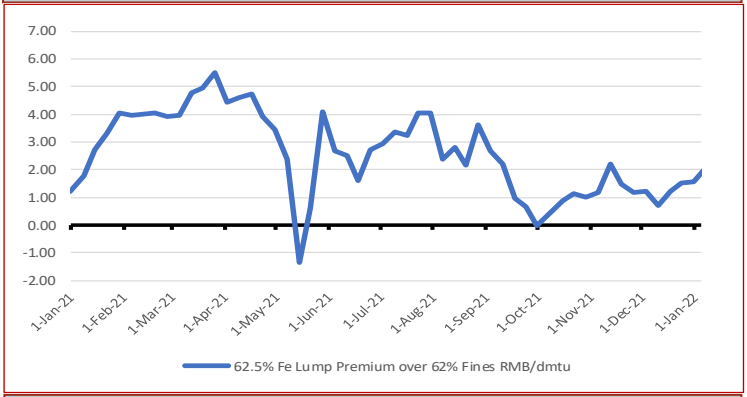
IRON ORE INDEX PREMIUMS/DISCOUNTS

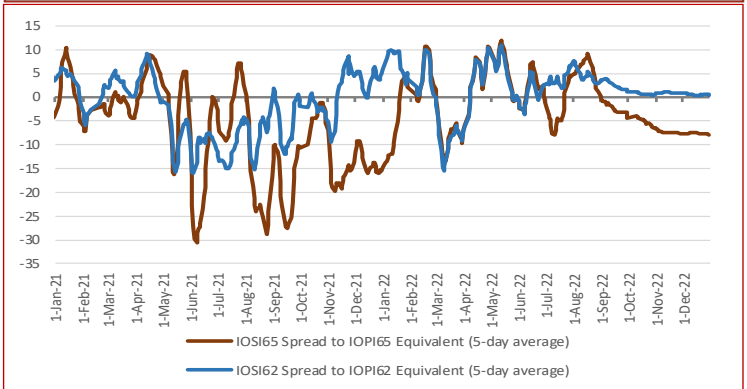




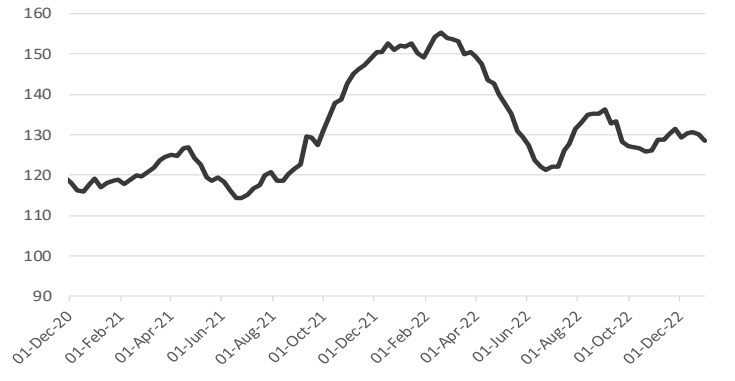
WEEKLY IRON ORE PORT STOCK LUMP PREMIUM (RMB/DMTU)



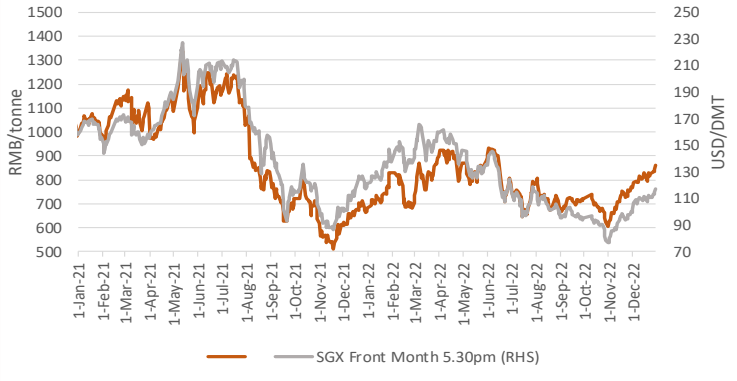
IRON ORE SEABORNE TO PORT STOCK PRICE SPREADS (USD/DMT)



TOTAL IRON ORE INVENTORIES AT CHINA PORTS (MILLION TONNES)



FUTURE TRADING—FRONT MONTH CLOSING PRICE



IRON ORE PORT INVENTORIES (MILLION TONNES)

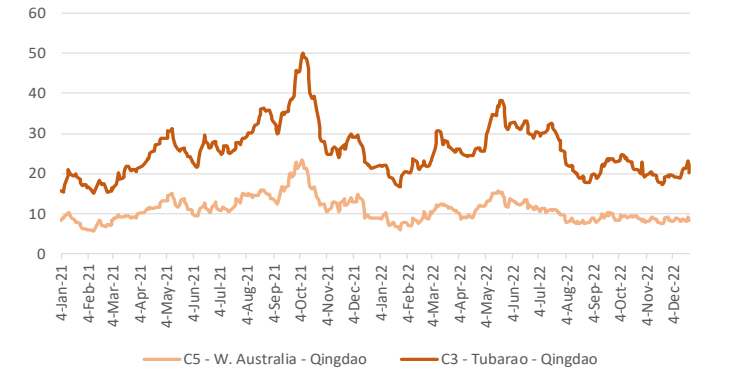
Week Ending Dec 30th, 2022

Province	This week	Change %	Low [2]	High [2]
Jingtang	12.44	-1.66%	9.84	19.20
Qingdao	25.34	-0.78%	9.41	26.24
Caofeidian	11.58	-5.31%	9.05	16.29
Tianjin	9.98	5.83%	7.14	12.97
Rizhao	17.33	-1.76%	9.44	19.26
Total (35 Ports)	128.47	-1.18%	98.80	155.39

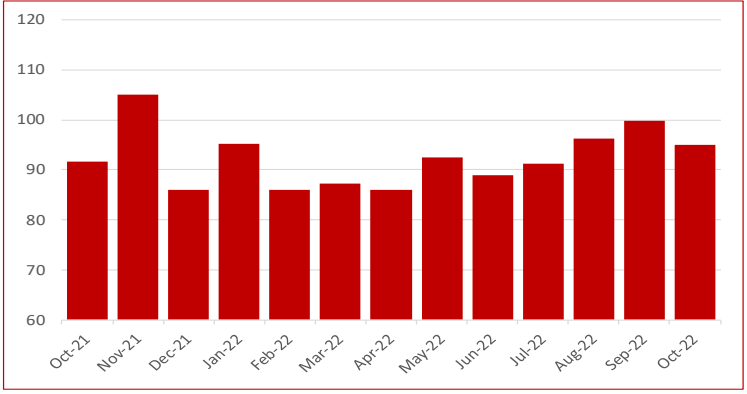
IRON ORE FUTURES CONTRACTS

	DCE (RMB/WMT)			SGX (USD/DMT)		
Closing Date	Dec 30th, 3pm close			Dec 30th, 5:30pm		
Contract	I2305	Change	Change %	Jan. 23	Change	Change %
Closing Price	863.00	18.00	2.13%	117.30	2.50	2.18%
Vol traded ('000 lots)	75.24	-1.43	-1.87%	2.17	-2.56	-54.10%
Open positions ('000 lots)	88.16	-0.37	-0.42%	26.24	-1.51	-5.43%
Day Low	842.5	11.50	1.38%	114.55	1.60	1.42%
Day High	867.5	20.00	2.36%	117.35	2.30	2.00%

DRY BULK FREIGHT RATES (USD/MT)



TOTAL CHINA IRON ORE IMPORT VOLUMES (MILLION TONNES)



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STEEL SPOT MARKET PRICES—CHINA

Steel Spot Market RMB/tonne

Product	2022/12/30	Change	Change %
ReBar HRB400 ϕ18mm	4,100	60	1.49%
Wirerod Q300 ϕ6.5mm	4,360	60	1.40%
HRC Q235/SS400 5.5mm*1500*C	4,180	100	2.45%
CRC SPCC/ST12 1.0mm*1250*2500	4,550	-30	-0.66%
Medium & Heavy Plate Q235B 20mm	4,130	-20	-0.48%
GI ST02Z 1.0mm*1000*C	4,790	-30	-0.62%
Colour Coated Plate	7,530	0	0.00%

CHINESE STEEL MILL PROFITABILITY

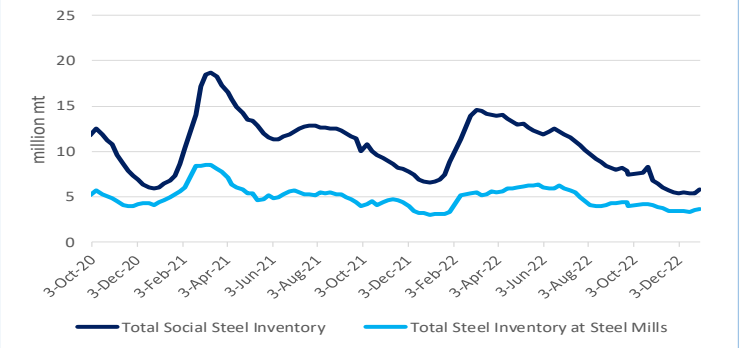
SMM Tracking of Steel Mill P&L - Rebar and Hot-rolled Coil (RMB/tonne)

Category	Price	Change (WoW)	Note
MMi (Fe 62%), USD/mt exluding tax	113.35	1	Mmi CFR Equivalent index for 1st Feb
Coke	3,180	0	2nd grade met coke, Tangshan, incl. tax
Steel Scrap	3,120	80	steel scrap (6mm) in Zhangjiagang, exl. tax
Billet Cost	3,808	25	Q234, incl. tax
Rebar cost - Blast furnace	4,337	28	calculated based on theoretical weight, incl. tax
Rebar profit - Blast furnace	-237	72	based on Shanghai prices, incl. tax
Hot-rolled coil cost - Blast furnace	4,295	26	based on actual weight, incl. tax
Hot-rolled coil proft - Blast furnace	-125	54	based on Shanghai prices, incl. tax

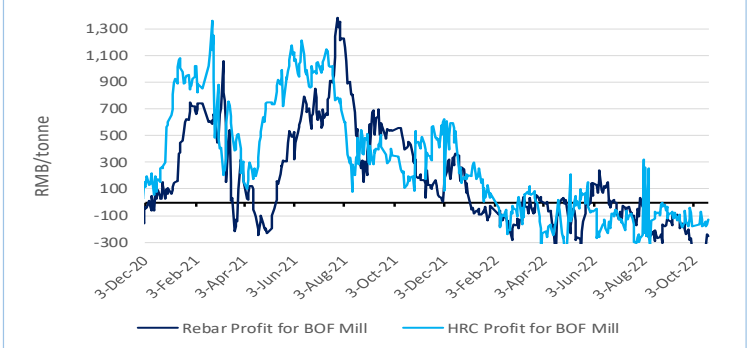
Note: 1. Costs in the table are caluclated based on todays market prices and facout our management, sales, financial and depreciations fe

2. The cost refers to average cost in the industry based on SMM's survey of small, medium and large mills in China

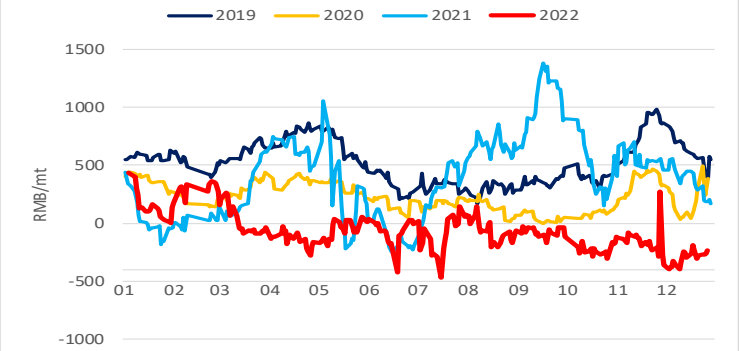
CHINESE STEEL INVENTORIES



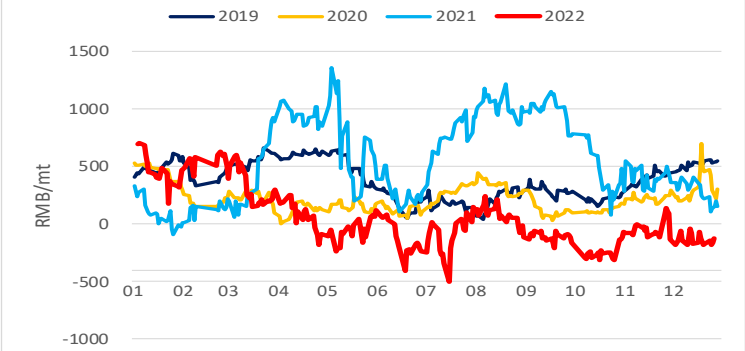
CHINESE STEEL MILL PROFITABILITY



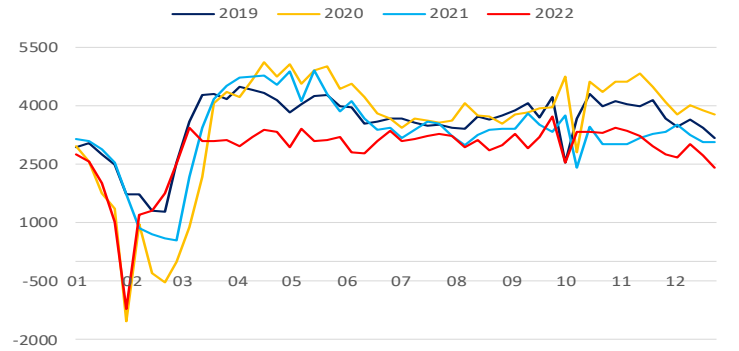
CHINESE STEEL MILL PROFITABILITY—Rebar



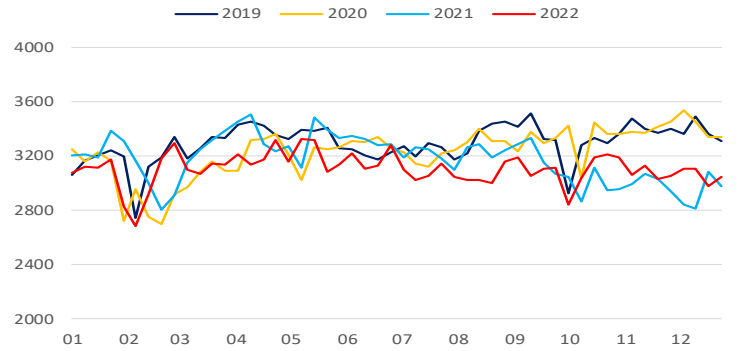
CHINESE STEEL MILL PROFITABILITY—Hot-rolled Coil



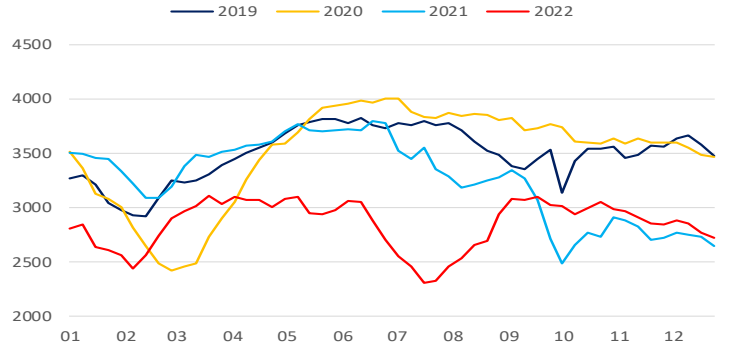
CHINESE STEEL CONSUMPTION—Rebar



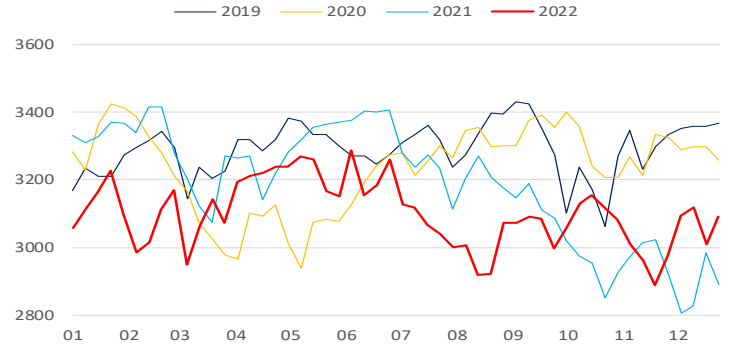
CHINESE STEEL CONSUMPTION—Hot-rolled Coil



CHINESE STEEL PRODUCTION—Rebar



CHINESE STEEL PRODUCTION—Hot-rolled Coil



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IRON ORE INDEX SPECIFICATIONS, COMPILATION RATIONALE AND DATA EXCLUSIONS

Iron Ore Index Specifications (Port and Seaborne)

	65% Fe Fines	62% Fe Fines	58% Fe Fines	62.5% Fe Lump
Fe %	65.00	62.00	58.00	62.50
Alumina %	1.40	2.25	2.25	1.50
Silica%	1.50	4.00	5.50	3.50
Phosphorus %	0.06	0.09	0.05	0.08
Sulphur %	0.01	0.02	0.02	0.02
Moisture %	8.00	8.00	9.00	4.00
Sizing	Granular size below 10mm for at least 90% of cargo; maximum of 40% below 150 micron			Size below 6.3mm max 15% Size above 31.15mm max 25%
Pricing Point	Qingdao Port (FOT and CFR respectively)			FOT Qingdao Port
Timing (Seaborne)	Loading within 4 weeks, Delivery within 8 weeks			Delivery within 2 weeks
Payment Terms	L/C at sight			L/C at sight or CAD

Iron Ore Index Compilation Rationale and Data Exclusions

MMi iron ore indices are compiled from data provided by companies that are part of the iron ore supply chain and involved in spot market transactions. The indices are calculated using detailed transaction-level data submitted to MMi by these companies. This data is normalised to the appropriate specifications and screened to remove outliers before volume-weighted average prices are calculated from the remaining core set of data.

For more details on MMi's iron ore methodology please download the guide published on our website at: www.mmiprices.com

Data Exclusions*

	62%	58%	65%
Port Index	0	0	0
Seaborne index	0		0
Lump Index 62.5	0		

* Number of price submissions for iron ore indices that were excluded from index calculations today as they were anomalous and could not be verified

IRON ORE DOMESTIC CONCENTRATE INDEX CALCULATION METHODOLOGY

The compilation method for price index generally refers to the compilation method of CPI price index and other price indices, breakdown the price data and calculate the average value according to a certain method, taking the vertical axis as the regional composite index (average of different grade index) and the horizontal axis as the grade composite index (average of different regional index), a total composite index for domestic ore can be output ultimately. The process system is also adopted in the calculation i.e. each sub-index can be obtained as well.

AVERAGE IRON ORE SPECIFICATIONS APPLIED FOR BRAND PRICE ASSESSMENTS

PORT STOCK BRANDS

Dec 30th, 2022	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.88%	2.35%	4.45%	0.055%	9.26%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines	61.64%	2.31%	3.87%	0.099%	9.45%
Newman Fines	62.49%	2.31%	3.98%	0.090%	7.72%
MAC Fines	60.43%	2.38%	4.64%	0.078%	7.57%
Jimblebar Blended Fines	60.87%	2.95%	4.53%	0.111%	7.32%
Carajas Fines	64.90%	1.39%	1.82%	0.080%	7.78%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	62.56%	1.58%	4.79%	0.091%	8.72%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

Dec 30th, 2022	Specifications applied for 58% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
SSF	56.49%	3.20%	6.19%	0.065%	9.18%
FMG Blended Fines	58.20%	2.43%	5.54%	0.057%	8.29%
Robe River	56.44%	3.16%	5.73%	0.042%	8.44%
Western Fines	57.88%	2.87%	7.50%	0.062%	7.45%
Atlas Fines	58.00%	1.85%	5.50%	0.090%	9.00%
Yandi	56.87%	1.58%	6.41%	0.042%	9.53%

SEABORNE BRANDS

	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.70%	2.30%	4.90%	0.055%	8.00%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines 62%	62.00%	2.60%	4.30%	0.090%	9.50%
Newman Fines	62.30%	2.40%	4.30%	0.095%	8.00%
MAC Fines	60.80%	2.35%	4.40%	0.075%	7.80%
Jimblebar Blended Fines	60.50%	3.10%	4.80%	0.120%	7.80%
Carajas Fines	65.10%	1.50%	1.70%	0.080%	8.50%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	63.00%	1.50%	5.00%	0.070%	7.00%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

BLOOMBERG TICKERS

PORT STOCK INDICES

	FOT Qingdao (RMB/wet tonne)	CFR Qingdao Equivalent (USD/dry tonne)
IOPI62	IRCNQ001	IRCNQ004
IOPI58	IRCNQ002	IRCNQ005
IOPI65	IRCNQ003	IRCNQ006
IOPLI62	IRCN0036	IRCN0037

SEABORNE INDICES

	CFR Qingdao (USD/DMT)
IOSI62	IRCN0034
IOSI65	IRCN0035

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